

3D Seismic Tomography with Simultaneous Hypocenter Estimation via Expectation–Maximization: Method and Application

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Abstract

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earthquake tomography; 3D velocity model

1 Introduction

Local-earthquake travel-time tomography uses picked seismic arrival times to estimate three-dimensional velocity structure and earthquake locations. Early work already formulated the inversion in terms of both source and medium parameters (Aki & Lee, 1976). An earthquake hypocenter is not an independently measured datum: its location is inferred from arrival times using an assumed velocity model, while velocity reconstruction in turn depends on the event locations (Kissling, 1988; Thurber, 1992). Treating catalog hypocenters as fixed inputs can therefore propagate location errors and dependencies on the models and procedures used to construct the catalog, particularly when those assumptions or location uncertainties are not documented. This coupling motivates estimating earthquake locations and velocity structure together.

Travel-time tomography remains useful for constructing and refining local velocity models (Fichtner et al., 2024). Such models can also provide starting points for waveform-based inversion, whose results may depend on the initial model (Virieux & Operto, 2009). There is therefore continued demand for practical methods that use arrival-time data to estimate three-dimensional structure without assuming accurately known earthquake locations.

Established joint-inversion methods iteratively update the velocity structure and an individual location estimate for each earthquake (Thurber, 1983). They differ in how those estimates are initialized: tomoDD uses initial event locations (Zhang & Thurber, 2003), whereas LOTOS determines them from arrival times and station coordinates in a one-dimensional reference model before three-dimensional inversion (Koulakov, 2009). In these formulations, locations enter velocity updates as individual estimates rather than as weighted alternatives. Probabilistic methods address different parts of this problem. Event-location methods can characterize alternative hypocenters given a prescribed velocity model (Lomax et al., 2000; Smith et al., 2022). Bayesian joint inversion has been demonstrated for both one-dimensional velocity structure (Ryberg & Haberland, 2019) and three-dimensional local-earthquake tomography using trans-dimensional Monte Carlo sampling (Piana Agostinetti et al., 2015); the latter jointly samples hypocenters and velocity structure without requiring catalog locations as starting coordinates. DeepGEM proposes generalized expectation–maximization for blind inversion (Gao et al., 2021),

while a recent three-dimensional variational double-difference method quantifies uncertainty in the velocity structure (Yang et al., 2025). These approaches differ in their treatment of location uncertainty and computational strategy; joint probabilistic 3D inversion without catalog starting locations is therefore not unique to EMTomo.

Against this background, we ask a narrower practical question: under otherwise matched settings, how does replacing one best-fitting hypocenter with several weighted location hypotheses affect three-dimensional travel-time inversion? We present EMTomo, an iterative framework that alternates between locating events in the current velocity model and updating the model by regularized inversion. It requires arrival-time picks associated with individual events, station coordinates, and an initial velocity model defined on the inversion grid; the initial model may be one-dimensional or three-dimensional. The inversion uses one selected phase (P or S) at a time and does not require catalog hypocenter coordinates. EMTomo uses differences between arrivals at pairs of stations for the same event, which eliminate that event's origin time. This is distinct from tomoDD's double-difference formulation, which compares arrival times from pairs of events at common stations (Zhang & Thurber, 2003). Candidate locations are selected from the within-event misfit and can contribute to the model update with weights. We compare single-location and multiple-hypothesis updates within EMTomo across a series of synthetic three-dimensional checkerboard tests, spherical anomalies, and a more natural tomography-derived velocity model, using both idealized and real event-station geometries. This within-method comparison isolates the effect of retaining multiple location hypotheses. We also use tomoDD as an established external reference to put reconstruction quality in the context of a widely used joint-inversion workflow, rather than to attribute differences between the two methods solely to the handling of location uncertainty. Both methods receive the same synthetic arrival-time picks, referenced to the earliest station arrival for each event; tomoDD additionally receives initial hypocenters obtained by perturbing the true synthetic locations, whereas EMTomo does not use initial event coordinates. Because the methods differ in their inputs, velocity parameterization, and interpolation, the comparison is intended to characterize their behavior in these specified settings rather than claim a universal resolution advantage.

[TODO: Before submission: specify how candidate-location weights relate to the proba-

bilistic model and whether the implemented iteration satisfies EM assumptions; add a paper roadmap once the experiments are finalized. In Experiments, document the independent PyKonal forward solver used to generate synthetic arrival times, the skfmm solver used by EMTomo, the chosen tomography-derived model and real source–receiver geometry, the checkerboard block sizes and spherical anomalies, and the shared per-event earliest-arrival reference, the noise applied to tomoDD’s true synthetic starting hypocenters, and whether tomoDD receives explicitly computed inter-event differential times or forms them internally. Describe the common arrival-time picks, differing inputs and initialization, grids, regularization, and runtime; report that EMTomo uses nearest-neighbor interpolation in the tests whereas tomoDD uses trilinear interpolation, and interpret resolution differences with this limitation in mind. If the Python implementation is released, provide a stable repository link and a code-availability statement. In Discussion, note that joint P/S inversion and time-lapse applications are outside the scope of this study.]

2 Method

2.1 Forward Problem: Ray-Based Travel-Time Computation

[TODO: Ray tracing scheme, eikonal solver / shooting. Reference to initial velocity model parameterization. – anisotropy is not included in our method]

The travel time τ_n^{ijk} from a candidate hypocenter cell (i, j, k) to station n is a functional of the 3D slowness field $S(\mathbf{x})$, discretized into L cells. The *sensitivity matrix* $\mathbf{G}$ relates perturbations in slowness to perturbations in travel time:

$$G_{n\ell}^{ijk} = \frac{\partial \tau_n^{ijk}}{\partial S_\ell}, \quad (1)$$

where S_ℓ is the slowness of cell ℓ and $G_{n\ell}^{ijk}$ equals the ray-path length through cell ℓ for the ray connecting (i, j, k) to station n . [TODO: Add eikonal / shooting reference here.]

Algorithm 1 EM-based 3D seismic tomography

Require: Arrival times $\{t_{ni}\}$; initial slowness model $\mathbf{s}^{(0)}$

Ensure: Updated slowness model $\mathbf{s}^*$; hypocenter set $\{(i^*, j^*, k^*)_e\}$

1: $k \leftarrow 0$

2: **repeat**

3: **E-step:** for each event e , evaluate Φ_{ijk} and locate $(i^*, j^*, k^*)_e \leftarrow \arg \min_{ijk} \Phi_{ijk}$

4: **M-step:** assemble $\tilde{\mathbf{G}}$ and $\mathbf{r}$; solve the regularized inverse problem for $\Delta \mathbf{s}^{(k)}$; update $\mathbf{s}^{(k+1)} \leftarrow \mathbf{s}^{(k)} + \Delta \mathbf{s}^{(k)}$

5: $k \leftarrow k + 1$

6: **until** $\|\Delta \mathbf{s}^{(k)}\| < \varepsilon$ **or** $k = k_{\max}$

7: **return** $\mathbf{s}^{(k)}$, hypocenter estimates

2.2 Probabilistic Formulation

[TODO: Observed data: arrival times $\{t_{ij}\}$.

Model parameters: velocity field $\mathbf{m}$.

Latent variables: hypocenter locations $\{\mathbf{h}_i\}$ and origin times.

Likelihood, prior, and complete-data log-likelihood.]

2.3 EM Algorithm

[TODO: Connecting text: formal EM interpretation — complete data, incomplete data,

Q-function. Cite Dempster et al. (1977).]

The full algorithm alternates between the two steps above until convergence (Algorithm 1).

2.3.1 E-step: Hypocenter Estimation

[TODO: Intro sentence: given fixed $\mathbf{s}^{(k)}$, find the best-fit hypocenter for each event.]

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